INCH-POUND

MIL-DTL-55302/170B
w/AMENDMENT 1
8 February 2010
SUPERSEDING
MIL-DTL-55302/170B
13 May 2005

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: LOW MATING FORCE, FEMALE BRUSH, RIGHT ANGLE, POLARIZED, FOR MULTILAYERED PRINTED WIRING BOARD (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

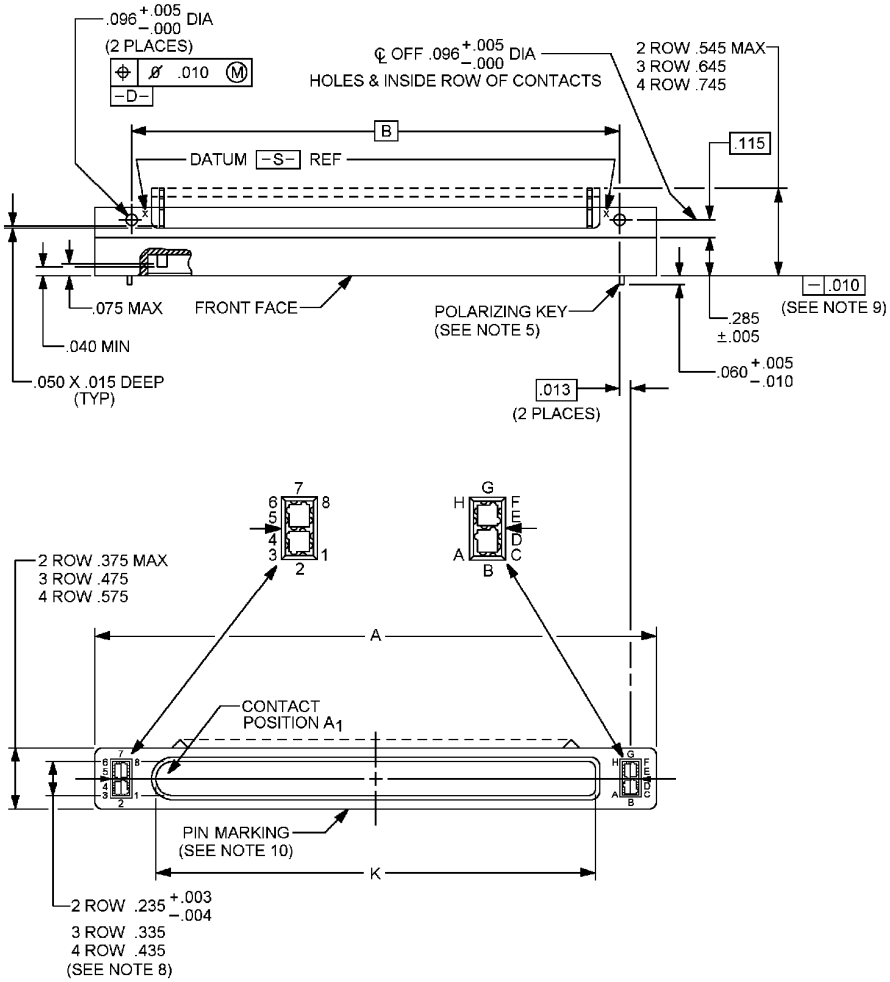
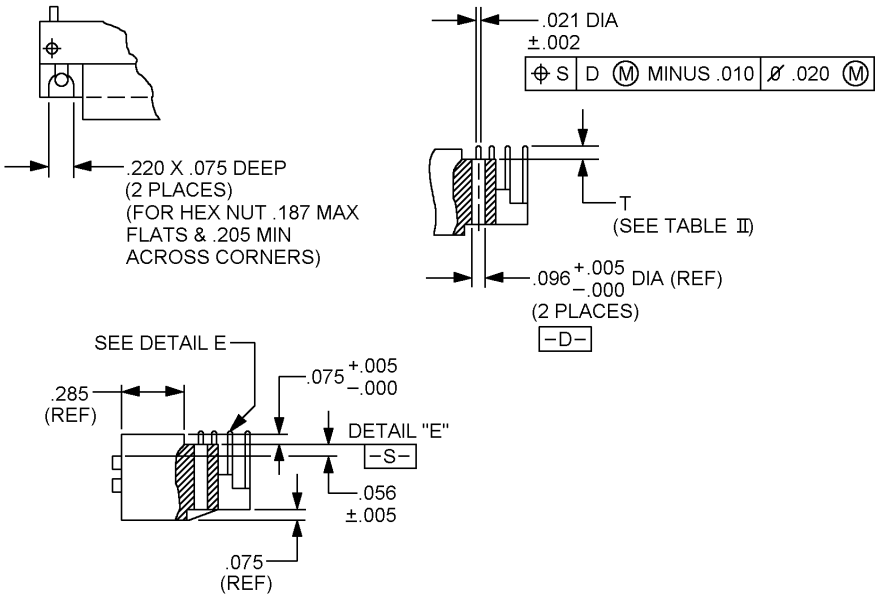



FIGURE 1. Connector, female brush, right angle for multilayered printed wiring boards .100 (2.54 mm) spacing.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .002 | 0.05 | .021 | 0.53 | .115 | 2.92 | .375 | 9.55 |
| .003 | 0.08 | .040 | 1.02 | .187 | 4.75 | .435 | 11.05 |
| .004 | 0.10 | .050 | 1.27 | .205 | 5.21 | .475 | 12.07 |
| .005 | 0.13 | .056 | 1.42 | .220 | 5.59 | .545 | 13.84 |
| .010 | 0.25 | .060 | 1.52 | .235 | 5.97 | .575 | 14.61 |
| .015 | 0.38 | .075 | 1.90 | .285 | 7.24 | .645 | 16.38 |
| .020 | 0.51 | .096 | 2.44 | .335 | 8.51 | .645 | 18.92 |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .010 (0.25 mm).
4. These connectors mate with connectors specified in MIL-DTL-55302/166, MIL-DTL-55302/167 and MIL-DTL-55302/168 and are primarily for use with single-sided, double-sided, or multilayered printed wiring boards.
5. The polarization feature, utilizing four M55302/78-02 polarizing keys which are not supplied with the connector, will provide 256 unique polarizing positions.
6. For dimensions A through K, see table I.
7. Contact rows are identified by molded letters on the front and rear face. Numbers indicating every fifth contact per row appear on rear face, except for arrangements consisting of 10 contacts per row which lack numerals. Up to 9 contact locations at the high numeral end of the contact row will not be identified.
8. Indicated dimension shall occur at least at one point within each of the following zones. The zones are defined as areas of the feature containing contact positions 1 through 5, 16 through 20, 46 through 50, 76 through 80, and the zone containing the highest five contact positions. The tolerance for the areas not described above shall be + .010 (0.25 mm) -.004 (0.10 mm).
9. Perfect form at MMC not required. Dimension for user installation purposes only.
10. Part or Identifying Number (PIN)

FIGURE 1. Connector, female brush, right angle for multilayered printed wiring boards .100 (2.54 mm) spacing – Continued.



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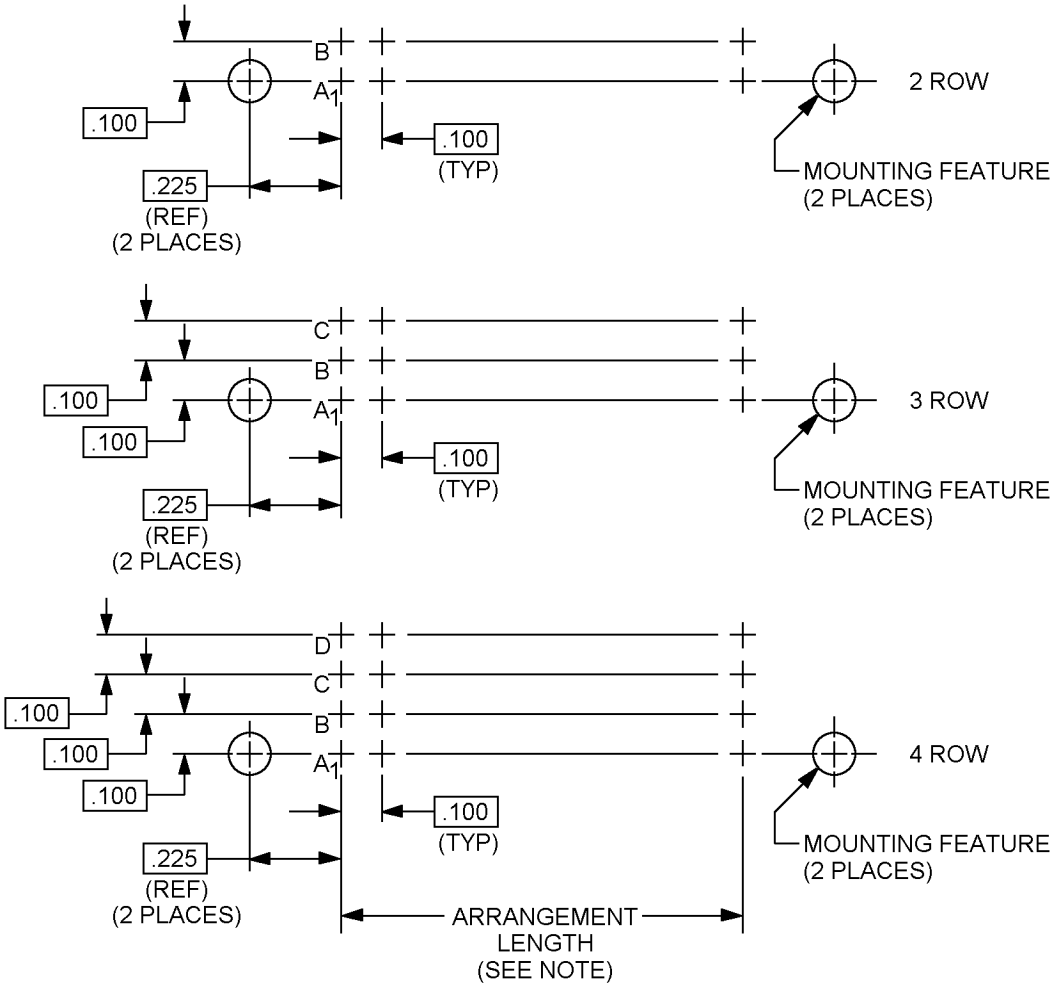
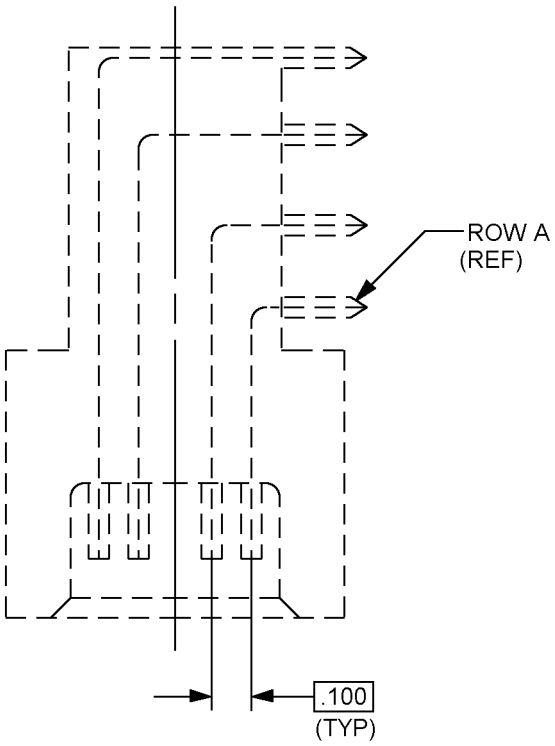



FIGURE 2. Insert arrangement for .100 (2.54 mm) contact spacing, female brush right angle connectors (PCB mounting side).



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| Inches | mm |
|---|---|
| .050 | 1.27 |
| .100 | 2.54 |
| .150 | 3.81 |
| .225 | 5.71 |

NOTE:
The arrangement length is determined by (N-1) x.100, where N is the number of contacts per row.
Example: For 60 contact row would be (60-1) x.100 = 5.90 inches.

FIGURE 2. Insert arrangement for .100 (2.54 mm) contact spacing, female brush right angle connectors (PCB mounting side) – Continued.



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REQUIREMENTS

Dimensions and configuration: See figure 1 and tables I and II.

Contacts: Each contact shall contain 7 brush wires with a tolerance limit of ± 1. Each brush wire diameter shall be .0073 to .0085 inch (0.185 mm to 0.216 mm).

Material:

Body: Glass filled thermoplastic molding material in accordance with ASTM D5948, type GPT-15F.

Contacts: In accordance with MIL-DTL-55302, ASTM B187/B187M or ASTM B272.

Plating: The contact termination plating shall be in accordance with table III. Brush wire plating shall be gold in accordance with ASTM B488, type II, code C, class 1.27 over nickel in accordance with SAE-AMS-QQ-N-290. Cut ends or brush wires need not be plated.

Contact identification: See figure 1 (note 7) and figure 2.

Oversized pin exclusion: Not applicable.

Contact engagement and separation forces: Not applicable, see mating and unmating.

Mating and unmating: The maximum mating and unmating force shall be 1.5 ounces times the total number of contacts.

Contact current rating: The connector may have any combination of current flow and ambient temperature provided the contact or connector temperature does not exceed 125°C. The test current shall be 5 amperes for individually connected contacts and 3 amperes for series wired contacts.

Contact resistance: The average resistance of all contact pairs measured on row A contacts shall not exceed 0.016 ohm, and no individual pair shall have a resistance exceeding 0.020 ohm. To allow for increased terminal lengths, add 0.001 ohm for each contact row beyond row A. The point of measurement shall be the point of terminal exit from the printed wiring board.

Contact retention: Not applicable.

Vibration: Vibration shall be in accordance with MIL-DTL-55302 except test condition V, letter H shall apply. Duration shall be 4 hours in each of three mutually perpendicular directions, 12 hours total.

Dielectric withstanding voltage:

Sea level: 1,300 volts rms, 60 Hz, ac.

High altitude: 325 volts rms, 60 Hz, ac.

Extended contact life: Connector shall meet the performance requirements of MIL-DTL-55302, contact life, when subjected to 20,000 insertion and withdrawal cycles during qualification inspection. Locking screws shall not be attached during contact life cycling.

Insert arrangement: See figure 2.



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PIN: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled as follows:

Example: M55302/170 B 18 Y 2

Specification sheet number

Number of rows (indicator)
(see table I for row indicator)

Number of contacts per row
(see table I for contact indicator)

.150 ± .025 contact tail termination length
(see table II for indicator)

Gold plated contact tails
(see table III for indicator)

User instructions: Each shipment of connectors shall include, as a minimum, instructions for coupling/mounting bushing usage, polarization key usage and electrical probing information.

TABLE I. Indicators and dimensions.

| 2 Contact rows | | 3 Contact rows | | 4 Contact rows | | Dimensions 1/ | | |
|---|---|---|---|---|---|---|---|---|
| Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | A Max | B | + .010 K - .000 |
| A10 | 20 | B10 | 30 | C10 | 40 | 1.680 (42.67) | 1.350 (34.29) | 1.120 (28.45) |
| A11 | 22 | B11 | 33 | C11 | 44 | 1.780 (45.21) | 1.450 (36.83) | 1.220 (30.99) |
| A12 | 24 | B12 | 36 | C12 | 48 | 1.880 (47.75) | 1.550 (39.33) | 1.320 (33.53) |
| A13 | 26 | B13 | 39 | C13 | 52 | 1.980 (50.29) | 1.650 (41.91) | 1.420 (36.07) |
| A14 | 28 | B14 | 42 | C14 | 56 | 2.080 (52.83) | 1.750 (44.45) | 1.520 (38.61) |
| A15 | 30 | B15 | 45 | C15 | 60 | 2.180 (55.37) | 1.850 (46.99) | 1.620 (41.15) |
| A16 | 32 | B16 | 48 | C16 | 64 | 2.280 (57.91) | 1.950 (49.53) | 1.720 (43.69) |
| A17 | 34 | B17 | 51 | C17 | 68 | 2.380 (60.45) | 2.050 (52.07) | 1.820 (46.23) |
| A18 | 36 | B18 | 54 | C18 | 72 | 2.480 (62.99) | 2.150 (54.61) | 1.920 (48.77) |
| A19 | 38 | B19 | 57 | C19 | 76 | 2.580 (65.53) | 2.250 (57.15) | 2.020 (51.31) |
| A20 | 40 | B20 | 60 | C20 | 80 | 2.680 (68.07) | 2.350 (59.69) | 2.120 (53.85) |
| A21 | 42 | B21 | 63 | C21 | 84 | 2.780 (70.61) | 2.450 (62.23) | 2.220 (56.39) |
| A22 | 44 | B22 | 66 | C22 | 88 | 2.880 (73.15) | 2.550 (64.77) | 2.320 (58.93) |
| A23 | 46 | B23 | 69 | C23 | 92 | 2.980 (75.69) | 2.650 (67.31) | 2.420 (61.47) |
| A24 | 48 | B24 | 72 | C24 | 96 | 3.080 (78.23) | 2.750 (69.85) | 2.520 (64.01) |
| A25 | 50 | B25 | 75 | C25 | 100 | 3.180 (80.77) | 2.850 (72.39) | 2.620 (66.55) |

See footnotes at end of the table



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TABLE I. Indicators and dimensions - Continued.

| 2 Contact rows | | 3 Contact rows | | 4 Contact rows | | Dimensions 1/ | | |
|---|---|---|---|---|---|---|---|---|
| Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | A Max | B | + .010 K - .000 |
| A26 | 52 | B26 | 78 | C26 | 104 | 3.280 (83.31) | 2.950 (74.93) | 2.720 (69.01) |
| A27 | 54 | B27 | 81 | C27 | 108 | 3.380 (85.85) | 3.050 (77.47) | 2.820 (71.63) |
| A28 | 56 | B28 | 84 | C28 | 112 | 3.480 (88.39) | 3.150 (80.01) | 2.920 (74.17) |
| A29 | 58 | B29 | 87 | C29 | 116 | 3.580 (90.93) | 3.250 (82.55) | 3.020 (76.71) |
| A30 | 60 | B30 | 90 | C30 | 120 | 3.680 (93.47) | 3.350 (85.09) | 3.120 (79.25) |
| A31 | 62 | B31 | 93 | C31 | 124 | 3.780 (96.01) | 3.450 (87.63) | 3.220 (81.79) |
| A32 | 64 | B32 | 96 | C32 | 128 | 3.880 (98.55) | 3.550 (90.17) | 3.320 (84.33) |
| A33 | 66 | B33 | 99 | C33 | 132 | 3.980 (101.09) | 3.650 (92.17) | 3.420 (86.87) |
| A34 | 68 | B34 | 102 | C34 | 136 | 4.080 (103.63) | 3.750 (93.98) | 3.520 (89.41) |
| A35 | 70 | B35 | 105 | C35 | 140 | 4.180 (106.17) | 3.850 (97.79) | 3.620 (91.95) |
| A36 | 72 | B36 | 108 | C36 | 144 | 4.280 (108.71) | 3.950 (100.330 | 3.720 (94.49) |
| A37 | 74 | B37 | 111 | C37 | 148 | 4.380 (111.25) | 4.050 (102.870 | 3.820 (97.03) |
| A38 | 76 | B38 | 114 | C38 | 152 | 4.480 (113.79) | 4.150 (105.410 | 3.920 (99.57) |
| A39 | 78 | B39 | 117 | C39 | 156 | 4.580 (116.33) | 4.250 (107.95) | 4.020 (102.11) |
| A40 | 80 | B40 | 120 | C40 | 160 | 4.680 (118.87) | 4.350 (110.49) | 4.120 (104.65) |
| A42 | 82 | B41 | 123 | C41 | 164 | 4.780 (121.41) | 4.450 (113.03) | 4.220 (107.19) |
| A42 | 84 | B42 | 126 | C42 | 168 | 4.880 (123.95) | 4.550 (115.57) | 4.320 (109.73) |
| A43 | 86 | B43 | 129 | C43 | 172 | 4.980 (126.49) | 4.650 (118.11) | 4.420 (112.27) |
| A44 | 88 | B44 | 132 | C44 | 176 | 5.080 (129.03) | 4.750 (120.65) | 4.520 (114.81) |
| A45 | 90 | B45 | 135 | C45 | 180 | 5.180 (131.57) | 4.850 (123.19) | 4.620 (117.35) |
| A46 | 92 | B46 | 138 | C46 | 184 | 5.280 (134.11) | 4.950 (125.73) | 4.720 (119.89) |
| A47 | 94 | B47 | 141 | C47 | 188 | 5.380 (136.65) | 5.050 (128.27) | 4.820 (122.43) |
| A48 | 96 | B48 | 144 | C48 | 192 | 5.480 (139.19) | 5.150 (130.81) | 4.920 (124.97) |
| A49 | 98 | B49 | 147 | C49 | 196 | 5.580 (141.73) | 5.250 (133.35) | 5.020 (127.51) |
| A50 | 100 | B50 | 150 | C50 | 200 | 5.680 (144.27) | 5.350 (135.89) | 5.120 (130.05) |
| A51 | 102 | B51 | 153 | C51 | 204 | 5.780 (146.81) | 5.450 (138.43) | 5.220 (132.59) |
| A52 | 104 | B52 | 156 | C52 | 208 | 5.880 (149.35) | 5.550 (140.97) | 5.320 (135.13) |
| A53 | 106 | B53 | 159 | C53 | 212 | 5.980 (151.89) | 5.650 (143.51) | 5.420 (137.67) |
| A54 | 108 | B54 | 162 | C54 | 216 | 6.080 (154.43) | 5.750 (146.05) | 5.520 (140.21) |
| A55 | 110 | B55 | 165 | C55 | 220 | 6.180 (156.97) | 5.850 (148.59) | 5.620 (142.75) |

See footnotes at end of the table



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TABLE I. Indicators and dimensions - Continued.

| 2 Contact rows | | 3 Contact rows | | 4 Contact rows | | Dimensions 1/ | | |
|---|---|---|---|---|---|---|---|---|
| Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | A Max | B | + .010 K - .000 |
| A56 | 112 | B56 | 168 | C56 | 224 | 6.280 (159.51) | 5.950 (151.13) | 5.720 (145.29) |
| A57 | 114 | B57 | 171 | C57 | 228 | 6.380 (162.05) | 6.050 (153.67) | 5.820 (147.83) |
| A58 | 116 | B58 | 174 | C58 | 232 | 6.480 (164.59) | 6.150 (156.21) | 5.920 (150.37) |
| A59 | 118 | B59 | 177 | C59 | 236 | 6.580 (167.13) | 6.250 (158.75) | 6.020 (152.91) |
| A60 | 120 | B60 | 180 | C60 | 240 | 6.680 (169.67) | 6.350 (161.29) | 6.120 (155.45) |
| A61 | 122 | B61 | 183 | C61 | 244 | 6.780 (172.21) | 6.450 (163.83) | 6.220 (157.99) |
| A62 | 124 | B62 | 186 | C62 | 248 | 6.880 (174.75) | 6.550 (166.37) | 6.320 (160.53) |
| A63 | 126 | B63 | 189 | C63 | 252 | 6.980 (177.29) | 6.650 (168.91) | 6.420 (163.07) |
| A64 | 128 | B64 | 192 | C64 | 256 | 7.080 (179.83) | 6.750 (171.45) | 6.520 (165.61) |
| A65 | 130 | B65 | 195 | C65 | 260 | 7.180 (182.37) | 6.850 (173.99) | 6.620 (168.15) |
| A66 | 132 | B66 | 198 | C66 | 264 | 7.280 (184.91) | 6.950 (176.53) | 6.720 (170.69) |
| A67 | 134 | B67 | 201 | C67 | 268 | 7.380 (187.45) | 7.050 (179.07) | 6.820 (173.23) |
| A68 | 136 | B68 | 204 | C68 | 272 | 7.480 (189.99) | 7.150 (181.61) | 6.920 (175.77) |
| A69 | 138 | B69 | 207 | C69 | 276 | 7.580 (192.53) | 7.250 (184.15) | 7.020 (178.31) |
| A70 | 140 | B70 | 210 | C70 | 280 | 7.680 (195.07) | 7.350 (186.69) | 7.120 (180.85) |
| A71 | 142 | B71 | 213 | C71 | 284 | 7.780 (197.61) | 7.450 (189.23) | 7.220 (183.39) |
| A72 | 144 | B72 | 216 | C72 | 288 | 7.880 (200.15) | 7.550 (191.77) | 7.320 (185.93) |
| A73 | 146 | B73 | 219 | C73 | 292 | 7.980 (202.69) | 7.650 (194.31) | 7.420 (188.47) |
| A74 | 148 | B74 | 222 | C74 | 296 | 8.080 (205.23) | 7.750 (196.85) | 7.520 (191.01) |
| A75 | 150 | B75 | 225 | C75 | 300 | 8.180 (207.77) | 7.850 (199.39) | 7.620 (193.55) |
| A76 | 152 | B76 | 228 | C76 | 304 | 8.280 (210.31) | 7.950 (201.93) | 7.720 (196.01) |
| A77 | 154 | B77 | 231 | C77 | 308 | 8.380 (212.85) | 8.050 (204.47) | 7.820 (198.63) |
| A78 | 156 | B78 | 234 | C78 | 312 | 8.480 (215.39) | 8.150 (207.01) | 7.920 (201.17) |
| A79 | 158 | B79 | 237 | C79 | 316 | 8.580 (217.93) | 8.250 (209.55) | 8.020 (203.71) |
| A80 | 160 | B80 | 240 | C80 | 320 | 8.680 (220.47) | 8.350 (212.09) | 8.120 (206.25) |
| A81 | 162 | B81 | 243 | C81 | 324 | 8.780 (223.01) | 8.450 (214.63) | 8.220 (208.79) |
| A82 | 164 | B82 | 246 | C82 | 328 | 8.880 (225.55) | 8.550 (217.17) | 8.320 (211.33) |
| A83 | 166 | B83 | 249 | C83 | 332 | 8.980 (228.09) | 8.650 (219.71) | 8.420 (213.87) |
| A84 | 168 | B84 | 252 | C84 | 336 | 9.080 (230.63) | 8.750 (222.25) | 8.520 (216.41) |
| A85 | 170 | B85 | 255 | C85 | 340 | 9.180 (233.17) | 8.850 (224.79) | 8.620 (218.95) |

See footnotes at end of table.



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TABLE I. Indicators and dimensions - Continued.

| 2 Contact rows | | 3 Contact rows | | 4 Contact rows | | Dimensions 1/ | | |
|---|---|---|---|---|---|---|---|---|
| Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | Indicator 2/ | Total contacts | A Max | B | + .010 K - .000 |
| A86 | 172 | B86 | 258 | C86 | 344 | 9.280 (235.71) | 8.950 (227.33) | 8.720 (221.49) |
| A87 | 174 | B87 | 261 | C87 | 348 | 9.380 (238.25) | 9.050 (229.87) | 8.820 (224.03) |
| A88 | 176 | B88 | 264 | C88 | 352 | 9.480 (240.79) | 9.150 (232.41) | 8.920 (226.57) |
| A89 | 178 | B89 | 267 | C89 | 356 | 9.580 (243.33) | 9.250 (234.95) | 9.020 (229.11) |
| A90 | 180 | B90 | 270 | C90 | 360 | 9.680 (245.87) | 9.350 (237.49) | 9.120 (231.65) |
| A91 | 182 | B91 | 273 | C91 | 364 | 9.780 (248.41) | 9.450 (240.03) | 9.220 (234.19) |
| A92 | 184 | B92 | 276 | C92 | 368 | 9.880 (250.95) | 9.550 (242.57) | 9.320 (236.73) |
| A93 | 186 | B93 | 279 | C93 | 372 | 9.980 (253.49) | 9.650 (245.11) | 9.420 (239.27) |
| A94 | 188 | B94 | 282 | C94 | 376 | 10.080 (256.03) | 9.750 (247.65) | 9.520 (241.81) |
| A95 | 190 | B95 | 285 | C95 | 380 | 10.180 (258.57) | 9.850 (250.19) | 9.620 (244.35) |
| A96 | 192 | B96 | 288 | C96 | 384 | 10.280 (261.11) | 9.950 (252.73) | 9.720 (246.89) |
| A97 | 194 | B97 | 291 | C97 | 388 | 10.380 (263.65) | 10.050 (255.27) | 9.820 (249.43) |
| A98 | 196 | B98 | 294 | C98 | 392 | 10.480 (266.19) | 10.150 (257.81) | 9.920 (251.97) |
| A99 | 198 | B99 | 297 | C99 | 396 | 10.580 (268.73) | 10.250 (260.35) | 10.020 (254.51) |
| A00 | 200 | B00 | 300 | C00 | 400 | 10.680 (271.27) | 10.350 (262.89) | 10.120 (257.05) |

1/ Dimensions are in inches. Metric equivalents are based on 1 inch = 25.4 mm.
2/ Indicators consist of, a letter designating number of contact rows and two digits indicating number of contacts per row. Indicators A00, B00, and C00 denotes 100 contacts per row.

TABLE II. Contact tail termination length.

| Indicator | T dimension |
|---|---|
| Z | .120 ± .025 |
| Y | .150 ± .025 |
| X | .300 ± .025 |

TABLE III. Contact tail plating.

| Indicator | Plating |
|---|---|
| 1 | Tin lead in accordance with SAE-AMS-P-81728, 50 to 70 percent tin, .0001 minimum thick over copper plating |
| 2 | Gold in accordance with ASTM B488, type II, code C, class 0.51 over nickel in accordance with SAE-AMS-QQ-N-290 |



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/166
MIL-DTL-55302/167
MIL-DTL-55302/168
ASTM B187/B187M
ASTM B272
ASTM B488
ASTM D5948
SAE-AMS-P-81728
SAE-AMS-QQ-N-290

CONCLUDING MATERIAL

Custodians:
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2008-140)

Review activities:
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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